Project Title: Unsupervised Brain Tumor Detection Using BiGAN

Authors:

Name	Student ID
Ashim Chandra Das	20201042
Rajesh Mojumder	24166017

Submission Date:

25 June 2026

Advanced Training on Semiconductor and ICT Technology

BRAC University

Table of Contents

1. Introduction

1.1 Brief Introduction

Brain tumor detection is a critical challenge in medical imaging. Supervised approaches require large labeled datasets of tumor images, which are expensive and time-consuming to obtain.

This project builds an unsupervised anomaly detection system using a BiGAN (Bidirectional Generative Adversarial Network) trained exclusively on healthy brain MRI scans; no tumor images are needed during training.

- The selected domain is Medical Image Analysis: specifically, brain MRI tumor detection using unsupervised deep learning.
- The system learns the statistical distribution of healthy brain anatomy. At inference, images the model cannot reconstruct well are flagged as anomalous, indicating a potential tumor.
- Finally, the system is a four-stage ML pipeline (data ingestion, preprocessing, BiGAN training, evaluation) combined with a Streamlit prediction app served via Docker, providing real-time tumor detection with a visual heatmap showing the suspicious region.

1.2 Problem Statement

Manual analysis of brain MRI scans is time-intensive and prone to inter-observer variability. An automated system that can reliably flag suspicious regions supports radiologists and speeds up diagnosis.

- **Input:** A brain MRI image (JPG or PNG) uploaded by the user.
- **Output:** A binary prediction, Healthy or Tumor Detected, together with a pixel-level reconstruction error heatmap that highlights the anomalous region when a tumor is suspected.
- **Target users:** Medical professionals, researchers, and diagnostic decision-support systems in clinical or research settings.
- **Clinical relevance:** Early detection of brain tumors significantly improves patient outcomes. The unsupervised approach removes the dependency on large labeled tumor datasets, making the system practical in data-scarce medical environments.

1.3 Objectives

- Build a BiGAN model trained exclusively on healthy MRI images for unsupervised brain tumor detection.
- Detects brain tumors as reconstruction anomalies, without any tumor supervision, using anomaly score = 0.7xMSE + 0.3xSSIM error.

- Track all experiments (hyperparameters, losses, metrics, artifacts) using MLflow with DagsHub remote tracking.
- Provide a Streamlit prediction app that accepts an MRI upload and returns a diagnosis with a visual tumor region heatmap overlay.
- Containerize the prediction app using Docker for reproducible, platform-independent deployment.

2. Dataset and Preparation

2.1 Dataset Description

Dataset: Brain MRI dataset sourced from Kaggle Dataset.

Total samples: 1901 JPEG images; 988 healthy brain MRIs and 913 tumor brain MRIs.

- **Classes:** Two; Healthy (label 0) and Tumor (label 1). Labels are used only during evaluation; the training set contains healthy images only.
- **Input type:** Grayscale MRI images, converted to 128x128 pixels and normalised to [0, 1] during preprocessing.
- **Target variable:** Binary anomaly label (Healthy/Tumor), determined at evaluation time by thresholding the reconstruction-based anomaly score.

2.2 Data Access

Stage 01 of the pipeline downloads the dataset automatically from a Google Drive link using gdown. The archive is extracted to data/BRAIN MRI/ and the zip file is deleted immediately after extraction.

Raw data is not committed to the GitHub repository. Only the download script (src/tumor_detection/components/data_ingestion.py) and the configured source URL (configs/config.yaml) are version-controlled.

2.3 Preprocessing

Stage 02 applies the following preprocessing steps to every image:

- Convert to grayscale (single channel).
- Resize to 128x128 pixels using PIL.
- Normalise pixel values to the range [0, 1] by dividing by 255.
- Split: 80% of healthy images (790) form the training set; the remaining 20% healthy (198) plus 20% of tumor images (183) form the test set. Tumor images are never used during training.
- Save as NumPy arrays: X_train.npy, X_test.npy, y_test.npy; in artifacts/data_transformation/.

3. Methodology

3.1 Course Techniques Used

The following course techniques were applied in this project (see Table 1 below for file-level mapping):

Technique	Where used
Unsupervised deep learning (BiGAN/GAN)	model/BIGAN.py, components/model_training.py

Technique	Where used
MLflow experiment tracking	components/model_training.py, components/model_evaluation.py
Dockerized prediction app	Dockerfile
Streamlit prediction UI	app.py
Anomaly detection evaluation (ROC, PR curve, confusion matrix)	components/model_evaluation.py

3.2 Model Architecture

The model is a BiGAN with three jointly trained components (defined in model/BIGAN.py).
Encoder: 4x Conv2D blocks (32 to 64 to 128 to 256 filters, 4x4 kernel, stride=2) with channel-wise attention after each block, followed by Dense(512) to Dense(128) to produce a 128-dimensional latent code.

Generator: Dense to Reshape(8,8,256), then 4x Conv2DTranspose blocks (256 to 128 to 64 to 32 to 16) with residual connections and attention, ending with Conv2D + sigmoid to output a 128x128x1 image.

Discriminator: dual-branch architecture (image branch via Conv2D, latent branch via Dense) concatenated and classified as real or fake (image, latent) pairs using binary cross-entropy.

Loss:

- **Discriminator:** binary cross-entropy.
- **BiGAN:** binary cross-entropy.
- **Reconstruction:** 0.6xMSE + 0.3xMAE + 0.1xSSIM.

Optimiser: Adam lr=5e-5 (discriminator, BiGAN), lr=1e-4 (reconstruction).

Hyperparameters: latent_dim=128, img_size=128, batch_size=32.

3.3 Training Strategy

Training uses a two-phase strategy on 790 healthy images (batch size 32).

Phase 1 Pre-training (150 epochs): only the reconstruction model (encoder + generator) is trained with the combined pixel-level loss. This bootstraps the generator to produce recognisable reconstructions before adversarial training begins.

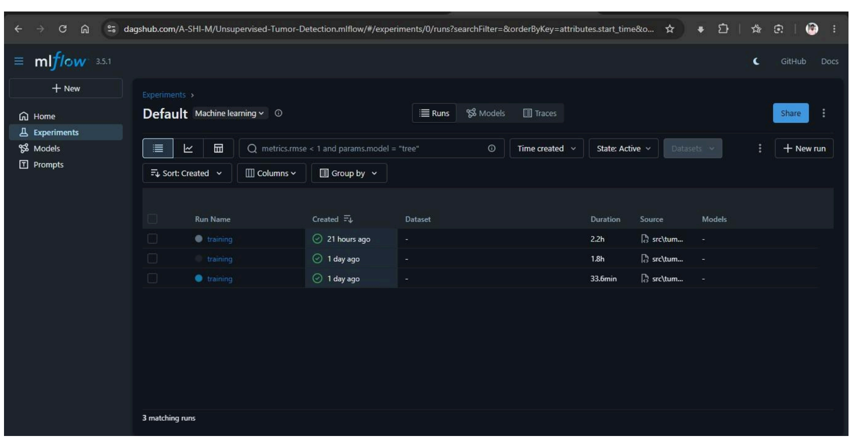
Phase 2 Joint training (500 epochs): the discriminator is updated every 5 epochs; the BiGAN (encoder + generator) is updated every epoch; the reconstruction model is updated 3x per epoch for stable convergence. After training, encoder.keras, generator.keras, and discriminator.keras are saved to both artifacts/model_training/ and trainedmodels/.

4. Experiments and Results

4.1 MLflow Tracking

All experiments are tracked on a DagsHub-hosted MLflow server (https://dagshub.com/A-SHI-M/Unsupervised-Tumor-Detection.mlflow). Training and evaluation are logged under a single MLflow run using a shared run_id saved to trainedmodels/mlflow_run_id.txt.

- **Parameters logged:** epochs, pretrain_epochs, batch_size, img_size, latent_dim, lr_discriminator, lr_bigan, lr_reconstruction, perceptual_loss.
- **Metrics logged:** pretrain_recon_loss (every 10 pre-train epochs), d_loss, g_loss, recon_loss (every 10 joint-training epochs), roc_auc, accuracy, recall, specificity, f1_score, optimal_threshold.
- **Artifacts saved:** reconstruction progress images (every 20/50 epochs), encoder/generator/discriminator model files, loss curve plot, evaluation plots (ROC curve, PR curve, anomaly score distribution, confusion matrix heatmap, metrics summary bar chart), metrics.json.
- **Three training runs were logged:** Two with VGG16 perceptual loss (ROC AUC 0.652, accuracy 62.2%) and one with pixel-level loss only (ROC AUC 0.876, accuracy 87.7%). The pixel-level run was selected as best because the unscaled VGG feature loss prevented pixel reconstruction learning.



4.2 Evaluation Results

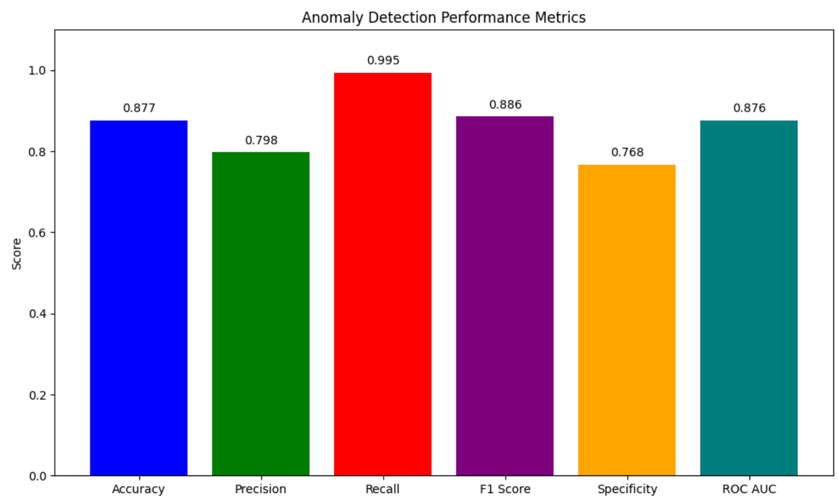
The retrained BiGAN (pixel-level loss, no VGG perceptual loss) was evaluated on 381 test images (198 healthy, 183 tumor) using reconstruction-based anomaly scoring (anomaly score = 0.7 x MSE + 0.3 x SSIM error). Results are summarised in Table 2 below.

Confusion matrix: TN=152, FP=46, FN=1, TP=182. The model misses only 1 tumor out of 183, demonstrating high clinical sensitivity. The optimal threshold (0.1202) was determined by Youden's J statistic on the ROC curve.

Metric	**Value**
ROC AUC	0.876
Accuracy	87.7%
Recall (Sensitivity)	99.5%
Specificity	76.8%
F1 Score	0.886
Average Precision	0.782
False Positives	46 / 198 healthy images
False Negatives	1 / 183 tumor images
Optimal Threshold	0.1202

Key finding: disabling the unscaled VGG16 perceptual loss (which dominated training at ~100,000x the scale of pixel MSE) and using only pixel-level loss raised ROC AUC from

0.652 to 0.876 and accuracy from 62.2% to 87.7%.



5. Prediction App and Docker

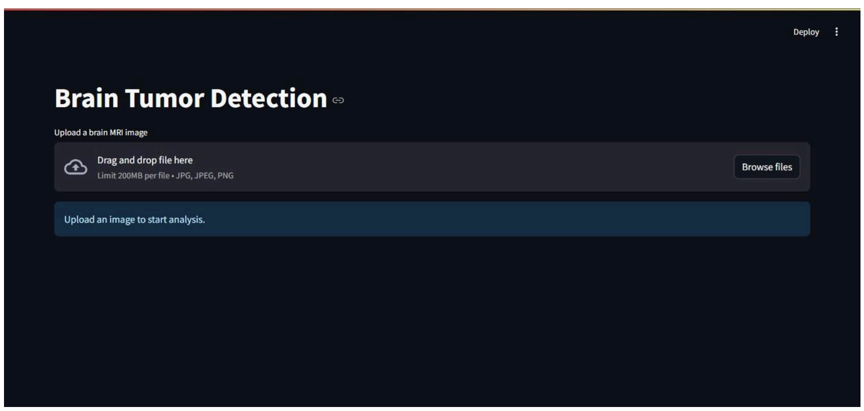
5.1 Prediction Pipeline

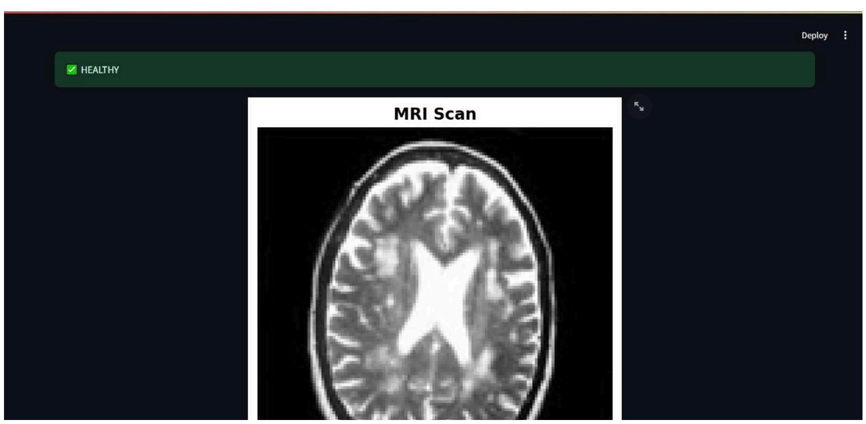
The prediction pipeline (app.py) operates as follows:

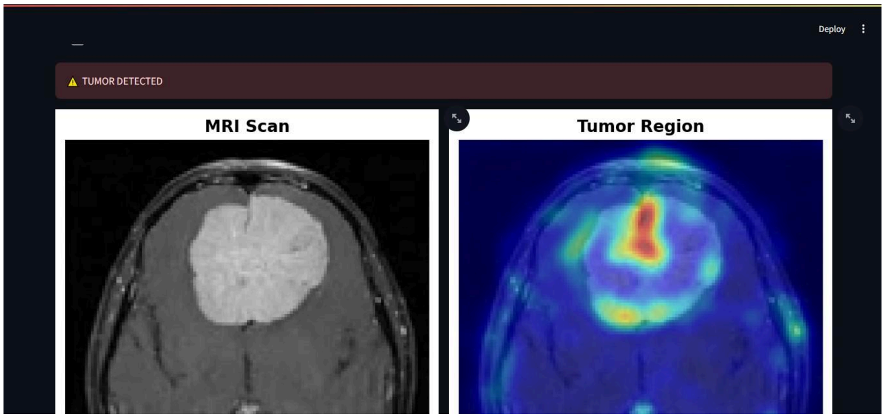
- The user uploads a brain MRI image (JPG or PNG) via the Streamlit file uploader.
- The image is converted to grayscale, resized to 128x128 pixels, and normalised to [0, 1].
- The trained encoder maps the image to a 128-dimensional latent code; the generator reconstructs the image from that code.
- Anomaly score = 0.7 x pixel MSE + 0.3 x SSIM error between the original and the reconstruction.
- If score >= 0.1202 (optimal threshold from metrics.json): Tumor Detected. For tumor predictions, a pixel-wise error map is computed, smoothed with a Gaussian filter (sigma=3), and overlaid on the original MRI as a jet-colourmap heatmap.

5.2 Prediction UI

The Streamlit app (app.py) shows a file uploader accepting JPG/PNG brain MRI images. After upload, inference runs automatically and the result is shown as a coloured banner: green (Healthy) or red (Tumor Detected). If Healthy: the original MRI image is displayed centred on the page. If Tumor: two panels are shown side by side — the original MRI (left) and the tumour region overlay (right), where a jet-colourmap heatmap (red/yellow = highest anomaly) is blended at 50% alpha on the MRI. The optimal threshold is loaded automatically from trainedmodels/metrics.json so the app always uses the latest calibrated value without code changes.

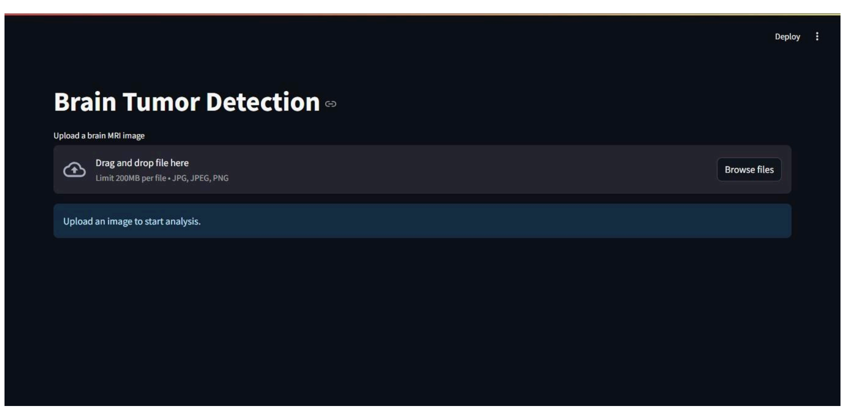






5.3 Docker Serving

- The prediction app is containerised using Docker (base image: python:3.12-slim, port 8501).
- The Dockerfile copies only the inference-required files: app.py, params.yaml, and the trainedmodels/ folder (models + metrics.json). Dependencies are installed from requirements.txt; the editable package install (-e .) is excluded since app.py does not import the tumor_detection package.
- **Build:** docker build -t tumor-detection:0.1 .
- **Run:** docker run -p 8501:8501 tumor-detection:0.1.
- The repository contains the Dockerfile and .dockerignore.



6. Repository and Reproducibility

6.1 Repository Structure

The repository is organised as follows:

- **src/tumor_detection/ :** pipeline package with four stages: data ingestion, data transformation, model training, model evaluation.
- **model/BIGAN.py:** BiGAN architecture (encoder, generator, discriminator).
- **configs/config.yaml:** pipeline paths.
- **params.yaml:** hyperparameters.
- **app.py:** Streamlit prediction UI
- **main.py**: full pipeline runner.
- Dockerfile, .dockerignore, requirements.txt, setup.py.

- **trainedmodels/:** encoder.keras, generator.keras, discriminator.keras, metrics.json, mlflow_run_id.txt.

6.2 GitHub Rules

The following are excluded from the repository via .gitignore:

- Raw dataset files (data/) - fetched at runtime by Stage 01.
- Virtual environment (venv/).
- Training artifacts (artifacts/) - generated by the pipeline.
- Credentials (.env file containing DagsHub token).

6.3 Reproducibility

A reviewer can reproduce the full pipeline from a fresh clone in the following order:

Step 1: Create Virtual Environment with python 3.12.10

Step 2: Install dependencies:

- pip install -r requirements.txt

Step 3: Get a DagsHub token:

- Create an account at dagshub.com.
- Go to User Settings -> Tokens -> Generate New Token.

Step 4: Create a .env file in the project root:

MLFLOW_TRACKING_URI=https://dagshub.com/A-SHI-M/Unsupervised-Tumor-Detection.mlflow

MLFLOW_TRACKING_USERNAME=dagshub-username

MLFLOW_TRACKING_PASSWORD=dagshub-token

Step 5: Run the full pipeline:

- python main.py (downloads data, transforms, trains BiGAN, evaluates)

Step 6: Inspect MLflow runs on DagsHub:

- https://dagshub.com/A-SHI-M/Unsupervised-Tumor-Detection.mlflow/

Step 7: Launch the prediction UI:

- streamlit run app.py

Step 8: Build and run the Docker container:

- Build Image: docker build -t tumor-detection:0.1
- Run Container: docker run -p 8501:8501 tumor-detection:0.1

7. Limitations and Future Work

7.1 Limitations

Small training dataset: only 790 healthy MRI images were used for training. A larger dataset would tighten the model's healthy distribution and reduce false positives.

Remaining false positives: 46 out of 198 healthy images are incorrectly flagged as tumors (~23%). Clinical deployment would require further calibration.

Binary classification only: the system outputs Healthy or Tumor without indicating tumor type, grade, or precise anatomical location.

Threshold sensitivity: the optimal threshold was calibrated on the current test split; it may need adjustment for images from different MRI scanners or acquisition protocols.

7.2 Future Improvements

Data augmentation: apply horizontal/vertical flips, rotations, and brightness jitter to expand the 790 training images without additional data collection.

Semi-supervised learning: incorporate a small number of labelled tumour images to tighten the decision boundary.

GAN-based data augmentation: use a conditional GAN to synthesise additional healthy MRI variants for training.

Improved anomaly localisation: replace the pixel-MSE heatmap with Grad-CAM or attention map visualisation for more interpretable region highlighting.

Clinical threshold calibration: validate the optimal threshold against radiologist-labelled ground truth from diverse MRI acquisition sources.

8. Conclusion

This project successfully built an unsupervised brain tumor detection system using a BiGAN trained exclusively on 790 healthy brain MRI images. Without ever observing tumor images during training, the final model achieves a ROC AUC of 0.876 and a recall of 99.5%, correctly identifying 182 out of 183 tumour cases (missing only 1). The system is deployed as a Streamlit web application containerised with Docker, providing real-time predictions with a visual tumour region heatmap generated from reconstruction error. A key experimental finding was that standard pixel-level reconstruction loss (0.6xMSE + 0.3xMAE + 0.1xSSIM) significantly outperforms VGG16 perceptual loss for this task. Unscaled VGG feature differences dominated the gradient at ~100,000x the scale of pixel MSE, preventing the model from learning pixel-level reconstruction quality. The team gained practical experience in unsupervised deep learning, GAN training dynamics, MLflow experiment tracking, and production-ready ML deployment with Docker and Streamlit.